

1 Methodology

1.1 Study Area

The study was conducted over Kyrgyzstan, a landlocked mountainous country in Central Asia located between approximately 39° N–43° N latitude and 69° E–80° E longitude. The terrain is dominated by the Tian Shan mountain system, with elevations ranging from approximately 400 m in the Fergana Valley to over 7,800 m at high alpine peaks. This extreme altitudinal gradient makes Kyrgyzstan highly sensitive to cryospheric change, and its snow and glacial resources are critical for downstream water supply across the Syr Darya and Amu Darya river basins, upon which millions of people in Central Asia depend.

1.2 Data Source

Snow Cover Data Snow cover data were obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS) combined Aqua and Terra 8-day maximum snow cover product (MOYD10A2, Collection 6.1), provided by the International Centre for Integrated Mountain Development (ICIMOD). The dataset covers the period from 2002 to 2023 and consists of 941 individual raster files, each representing an 8-day composite snow cover extent at approximately 500 m spatial resolution. Each image is a binary raster in which a pixel value of 1 indicates the presence of snow cover and 0 indicates its absence. The use of the combined Aqua-Terra product reduces cloud contamination compared to single-sensor products by taking the maximum snow extent observed across both satellites within each 8-day window. All files are provided in GeoTIFF format in the geographic coordinate system WGS84 (EPSG:4326). Individual images are identified by a Julian day-of-year naming convention (YYYYDDD), from which acquisition dates were programmatically extracted.

Digital Elevation Model Elevation data were derived from the ALOS World 3D 30 m Digital Elevation Model (AW3D30), provided by ICIMOD for the Kyrgyzstan domain. Two raster layers were used: (i) a raw elevation DEM at 30 m spatial resolution with values ranging from 395 m to 7,848 m, and (ii) a reclassified DEM in which continuous elevation values were categorised into four discrete elevation zones following ICIMOD standard classifications for Central Asian mountain regions. Both layers are provided in GeoTIFF format in WGS84 geographic coordinates. The four elevation zones and their corresponding altitude ranges are defined in Table 1.

Table 1: Elevation zone classification used in the analysis.

Zone	Elevation Range	Terrain Type
1	<1500m	Valley/Lowland
2	1500 - 3000 m	Montane
3	3000 - 4500 m	Sub-alpine
4	> 4500 m	Alpine/Nival

1.3 Data Pre-Processing

Pixel Area Estimation The area represented by each MODIS snow cover pixel was calculated to convert pixel counts into snow cover area (km^2). Given the nominal spatial resolution of approximately 500 m and the geographic coordinate system of the data, pixel dimensions in kilometres were estimated using a latitude correction applied at the centroid latitude of Kyrgyzstan.

$$A_{\text{pixel}} = W_{\text{deg}} \times H_{\text{deg}} \times \cos\left(41^\circ \times \frac{\pi}{180^\circ}\right) \times 111.0^2 \text{ km}^2 \quad (1)$$

This accounts for the convergence of meridians at mid-latitudes and yields a pixel area of approximately 0.134 km^2 , consistent with the expected resolution of the MOYD10A2 product at this latitude.

Spatial Co-registration The classified DEM (30 m resolution, $14,709 \times 39,709$ pixels) and the MODIS snow cover rasters ($1,068 \times 2,882$ pixels) were provided at substantially different spatial resolutions. To enable pixel-level correspondence between the two datasets, the classified DEM was resampled to match the spatial extent, resolution, and grid alignment of the MODIS snow cover data using nearest-neighbour interpolation via the rasterio reprojection module. Nearest-neighbour resampling was selected in preference to bilinear or cubic methods because the classified DEM contains discrete categorical values (integer classes 1–4), for which interpolation between class values would be physically meaningless. Following resampling, each pixel in the snow cover stack corresponds directly to a single elevation zone in the resampled classified DEM.

Seasonal Definition Each 8-day snow cover image was assigned to one of four seasons based on its acquisition date. Season boundaries were defined to reflect the hydroclimatic regime of Kyrgyzstan rather than standard meteorological seasons, following conventions used in ICIMOD cryosphere assessments for Central Asia:

Snow Cover Area Computation Snow Cover Area (SCA) was computed for each elevation zone for every 8-day acquisition period across the full study period (2002–2023). For each image, pixels flagged as nodata (value = 255) were masked and excluded from computation. Within each elevation zone, the number of snow-covered pixels (value = 1) was summed and multiplied by the estimated pixel area to yield SCA in km^2 . SCA was also expressed as a percentage of the total valid pixel area within each zone to facilitate comparison across zones of different sizes. This produced a time series of 941 SCA values per zone, forming the primary dataset for all subsequent analyses.

Temporal Aggregation The 8-day SCA time series was aggregated to two temporal scales for analysis:

Annual: Mean SCA across all 8-day periods within each calendar year, yielding 21 annual values per zone (2002 to 2022; 2023 was retained where a complete year was available).

Seasonal: Mean SCA across all 8-day periods within each season within each year, yielding up to four seasonal values per zone per year. Seasonal aggregation used the hydroclimatically informed season definitions described in section 1.3

Table 2: Season definitions used in the analysis..

Season	Months	Hydro-climatic Range
Winter	November - March	Full snow accumulation period; temperatures remain below freezing across most elevations
Spring	April - June	Primary snowmelt and runoff generation season
Summer	July - September	Snow-free at lower elevations; glacial melt dominant at high elevations
Autumn	October	Onset of re-accumulation at high elevations; transitional period

2 Trend Analysis

Long-term trends in annual SCA were assessed for each elevation zone using two complementary statistical approaches:

2.1 Linear Regression

Ordinary least squares (OLS) linear regression was applied to the annual SCA time series for each zone, with year as the independent variable and mean annual SCA (km^2) as the dependent variable. The regression slope quantifies the mean rate of change in SCA per year ($km^2/year$). Statistical significance was assessed using a two-tailed t-test at a significance level of $\alpha = 0.05$. The coefficient of

determination (R^2) was reported as a measure of the proportion of variance in annual SCA explained by the linear trend.

2.2 Mann-Kendall Trend Test and Sen’s Slope Estimator

Given that snow cover time series may not satisfy the normality assumption required by parametric tests, the non-parametric Mann-Kendall (MK) trend test was additionally applied to each annual SCA series. The MK test evaluates the monotonic trend in a time series without assuming any particular distributional form, making it well suited to hydrometeorological and cryospheric datasets. Kendall’s tau (τ) was reported as a standardised measure of trend strength, and statistical significance was again assessed at $\alpha = 0.05$. Sen’s slope estimator, defined as the median of all pairwise slopes between data points, was used as a robust non-parametric estimate of the magnitude of change ($km^2/year$). The Mann-Kendall test was implemented using the `pymannkendall` Python package.

Where results from OLS regression and the Mann-Kendall test were consistent in both direction and significance, greater confidence was placed in the identified trend.

2.3 Spatial Analysis

Two spatial products were derived to characterise the distribution of snow cover across Kyrgyzstan:

Mean Snow Cover Probability: The temporal mean of all 941 binary snow cover images was computed pixel-by-pixel, yielding a map in which each pixel value represents the proportion of 8-day periods during which snow cover was observed at that location over the full study period (2002–2023).

Snow Cover Duration (SCD): For each pixel, the total number of 8-day periods across all years in which snow cover was recorded was summed, producing a map of cumulative snow cover persistence. Higher SCD values indicate locations where snow cover is sustained for longer periods, typically corresponding to high-elevation terrain.

Both spatial products were computed using a single-pass accumulation approach to avoid loading the full image stack (≈ 10.8 GB) into memory simultaneously, instead accumulating running sums across sequentially read images.

2.4 Line Altitude Estimation

As a proxy for snow line altitude, the highest elevation zone in which at least 50% of valid pixels were snow-covered was identified for each 8-day acquisition period. The midpoint elevation of that zone was assigned as the approximate snow line altitude for that period. Annual mean snow line altitudes were then computed, and a linear trend was fitted to the resulting annual time series to assess whether the snow line showed a statistically significant upward or downward shift over the study period. A rising snow line indicates retreat of snow cover to higher elevations, consistent with warming temperatures.

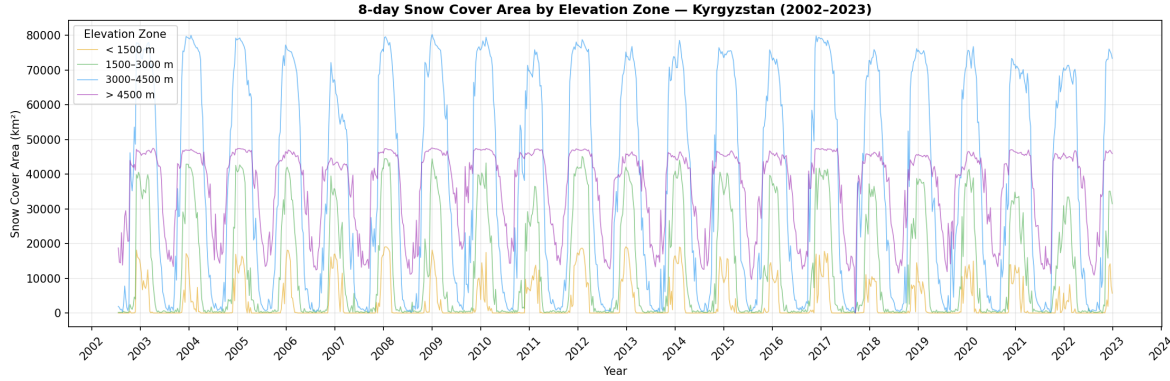


Figure 1: 8-day time series

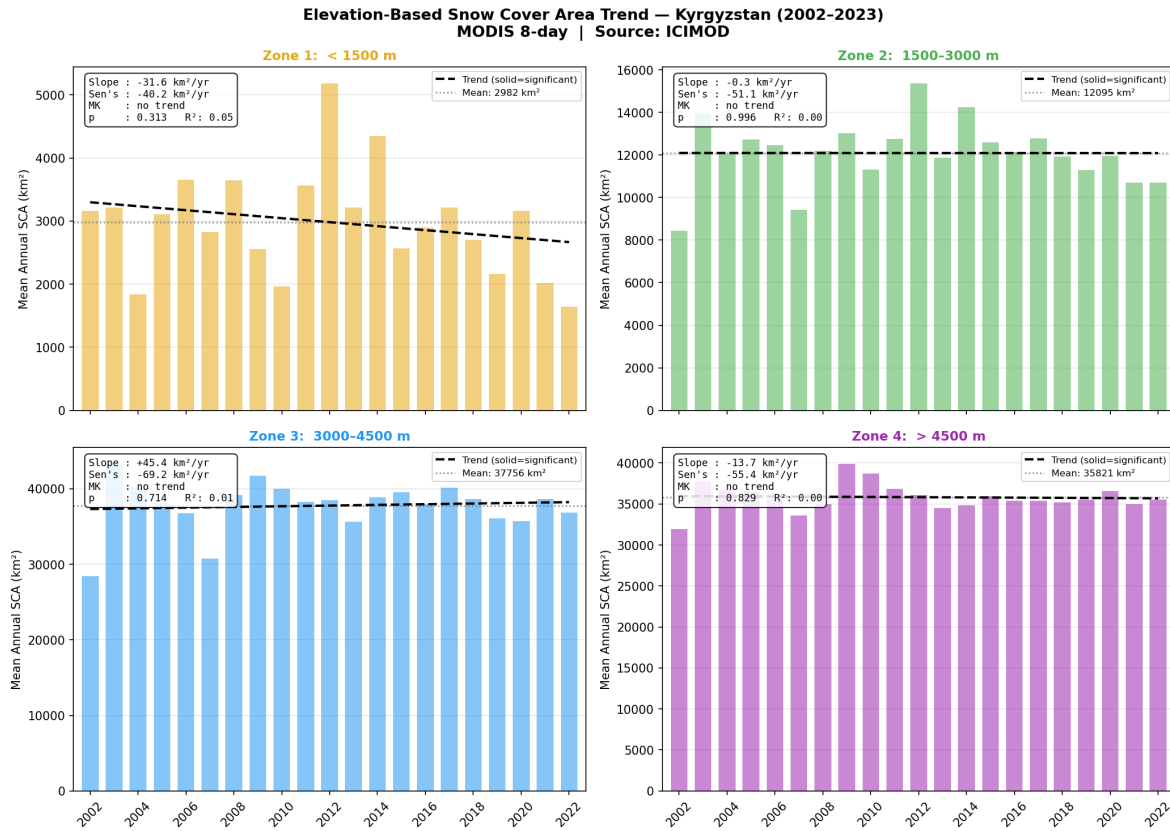


Figure 2: annual trend 4 panels

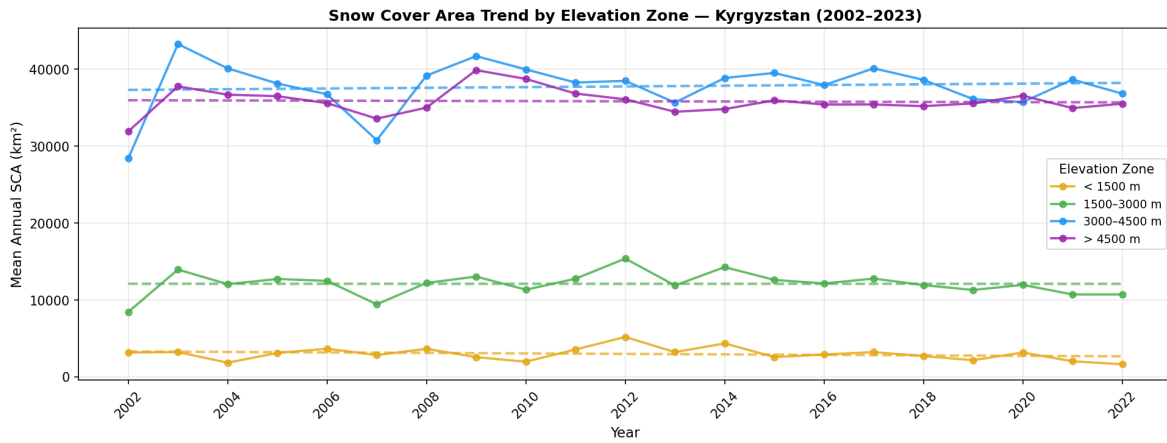


Figure 3: All Zones

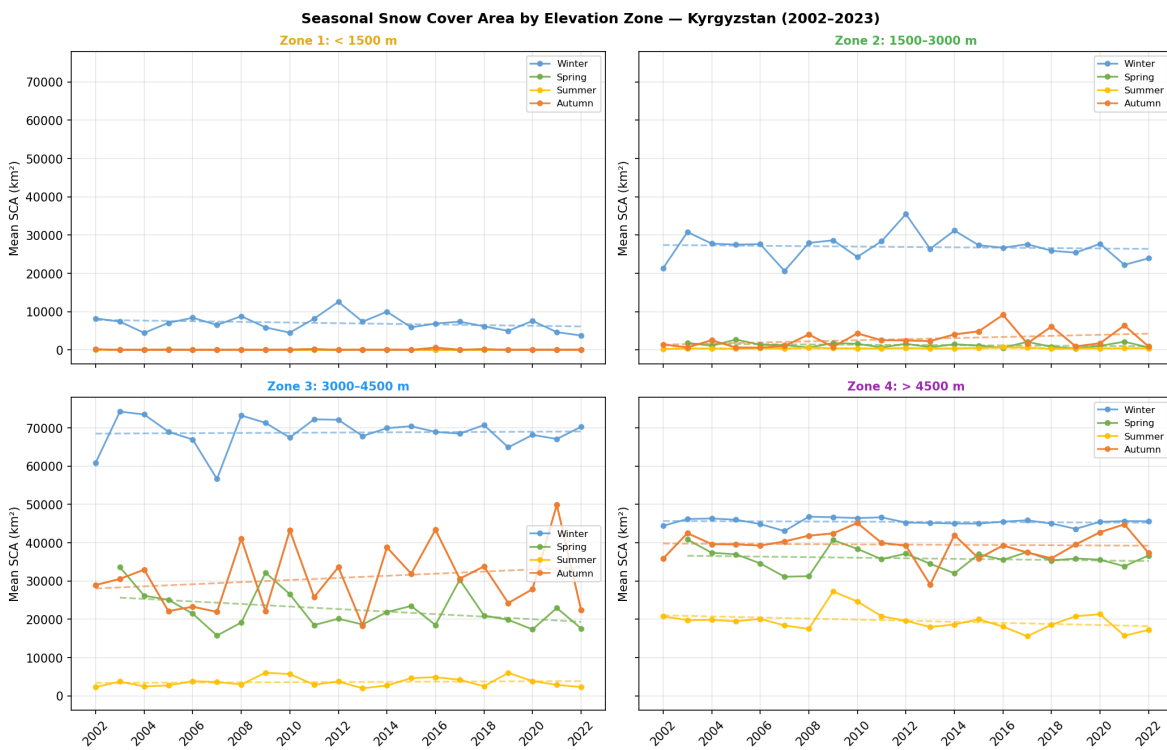


Figure 4: Seasonal Breakdown

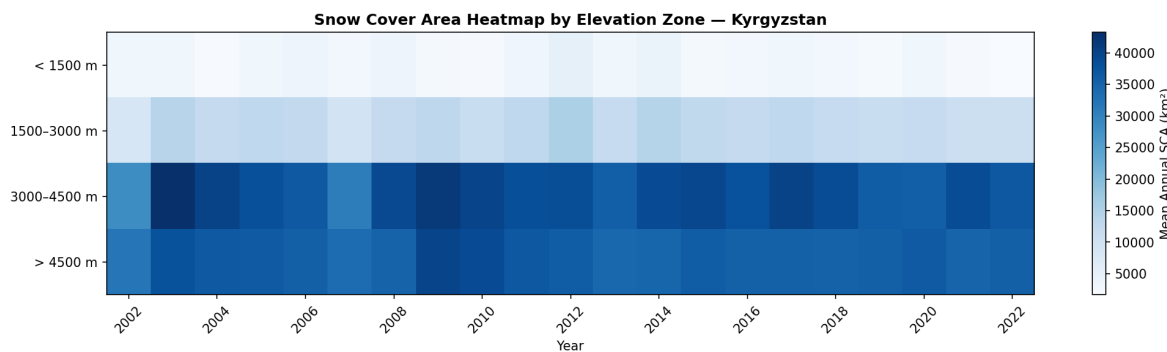


Figure 5: Heatmap

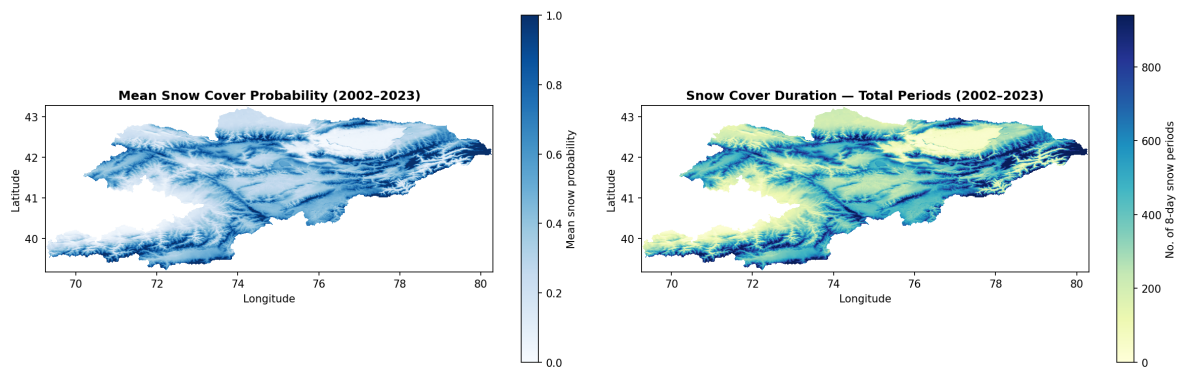


Figure 6: SCD

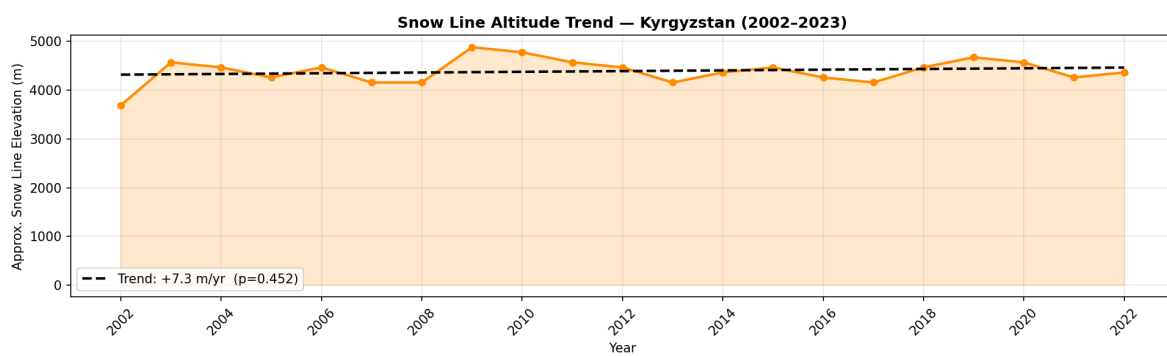


Figure 7: Snowline Trend